
Selective Neuro-Symbolic Reasoning with Structural Gating, Repair, and Fallback on ProofWriter

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Abstract

Large language models can translate natural-language reasoning problems into symbolic programs, but a program that parses may still be inadmissible for deterministic execution or semantically unfaithful to the source text. We study VGCF-2, our frozen selective neuro-symbolic pipeline for three-way ProofWriter classification under open-world semantics, using a local 9B open-weight model. It constrains an initial formalization, applies deterministic structural and executability checks followed by a forward solver, attempts one targeted repair when needed, and falls back to a strictly parsed chain-of-thought answer if the symbolic route remains unavailable. On a frozen, label-balanced Final sample of 300 questions, Full VGCF-2 obtains 90.33% accuracy. The two preregistered paired comparisons favor the full system over CoT-Refine by 8.67 percentage points (Holm-adjusted $p = 0.001371$) and over Gate+CoT by 2.67 points (Holm-adjusted $p = 0.038574$). These results support selective symbolic execution under this frozen configuration while identifying incomplete formalization coverage and imperfect semantic faithfulness as remaining limitations.

1 Introduction

ProofWriter asks whether a natural-language query is entailed, contradicted, or undetermined by a set of facts and forward rules [Tafjord et al., 2021]. The task is small enough for deterministic deduction once the text has been formalized, yet difficult for a language model because explicit negation, variable sharing, relation order, and open-world unknowns must all survive the translation. This creates a useful separation of responsibilities: a model can interpret language, while ordinary code can validate and solve the proposed representation.

Prior neuro-symbolic systems already translate text into solver inputs and use execution feedback for correction [Pan et al., 2023]. Other work keeps the reasoning inside the model but imposes symbolic plans and verification [Xu et al., 2024], or translates chain-of-thought steps into formulas for consistency checks [Feng et al., 2026]. A practical design question in this setting is how to handle a formalization that is not sufficiently reliable for symbolic execution. Rejecting it loses coverage; executing it may turn a translation defect into a confident answer; and unconstrained regeneration may introduce a new defect.

We investigate a narrow policy: validate a shared constrained formalization, repair it once using bounded diagnostics, execute only an admissible candidate, and otherwise use a shared natural-language fallback. The empirical question is whether this gate-repair-fallback composition improves task accuracy beyond natural-language self-refinement and beyond gating without repair in a small open-weight language model. The intervention is entirely at test time; it does not train or alter model weights.

This paper makes three contributions. First, we present a selective gate-repair-fallback pipeline that executes a compact symbolic representation only after deterministic structural and executability

checks, attempts one bounded repair otherwise, and retains a strictly parsed natural-language fallback when symbolic execution remains unavailable. Second, we conduct a controlled 7-method comparison on a frozen, label-balanced ProofWriter sample, sharing physical model responses wherever method designs overlap. Third, we evaluate the full system with two preregistered paired comparisons under a leakage-controlled model interface that exposes only the theory and question, and use secondary routing, repair, formalization, and semantic-fidelity diagnostics together with an explicitly post-hoc counterfactual analysis to characterize both the observed gains and the remaining limitations.

2 Background and Related Work

ProofWriter semantics. ProofWriter generates theories containing ground facts and implication rules, together with queries and proofs. Under the open-world setting used here, the label is True when the signed query is derivable, False only when its explicit complement is derivable, and Unknown when neither is derivable. Implications are applied forward; absence of a proof is not evidence for negation. This makes explicit negative atoms different from negation-as-failure [Tafjord et al., 2021].

Prompted and refined reasoning. Chain-of-thought prompting elicits intermediate natural-language reasoning without changing model parameters. It can expose an interpretable trace, but the trace is not guaranteed to be logically correct [Wei et al., 2022]. Self-Refine iterates between model-generated feedback and a revised output, without additional training. Its success depends on the quality of self-feedback, and repeated refinement can show diminishing returns [Madaan et al., 2023]. Our CoT-Refine comparison instantiates this natural-language review pattern once; it is not symbolic repair.

Neuro-symbolic reasoning and verification. Logic-LM maps a language problem to a task-specific symbolic formulation, invokes a deterministic solver, and uses solver feedback for self-refinement and falls back to chain-of-thought when a generated symbolic formulation is not solver-executable. It also reports execution rate alongside accuracy on executable cases, while noting that solver-valid syntax does not ensure a semantically correct translation [Pan et al., 2023]. VGCF-2 therefore shares Logic-LM’s broad formalize–solve–refine/fallback pattern, and our contribution is incremental with respect to that design. In our frozen pipeline, Gate+CoT and Full additionally require task-specific structural hard validity rather than solver executability alone, repair is limited to one bounded attempt, and the repair route is isolated by a preregistered paired comparison against an otherwise shared gate-plus-fallback method. SymbCoT instead uses a language model as translator, planner, solver, and verifier while retaining symbolic expressions alongside natural language [Xu et al., 2024]. VeriCoT autoformalizes chain-of-thought steps and tests logical consistency and grounding with external support, but it likewise cannot guarantee that a model-produced formula faithfully represents the corresponding sentence [Feng et al., 2026]. VGCF-2 occupies a narrower point in this design space: the gate checks structural and operational properties of one task-specific IR; it is not an LLM judge and not a semantic equivalence oracle.

3 Selective Gate–Repair–Fallback Reasoning

3.1 Model-input boundary and intermediate representation

Every method receives a model view constructed through one gateway containing exactly two fields: the natural-language theory and the question. Dataset answers, labels, proofs, generator representations, hidden rules, and metadata are evaluator-only. The gateway numbers source sentences so that validation and proof traces can retain auditable source identifiers.

The constrained formalizer returns one strict JSON object with facts, rules, and a query. An atom is signed positive or explicitly negative, has a lowercase predicate, and has one or two ordered arguments. Facts and the query are ground; variables occur only in rules. Each theory source identifier must be used exactly once and the question identifier may occur only on the query. The deterministic solver forward-chains ground substitutions to a fixed point. It returns False only when the explicit complement of the query is reached and reports inconsistency rather than guessing if both polarities are reached.

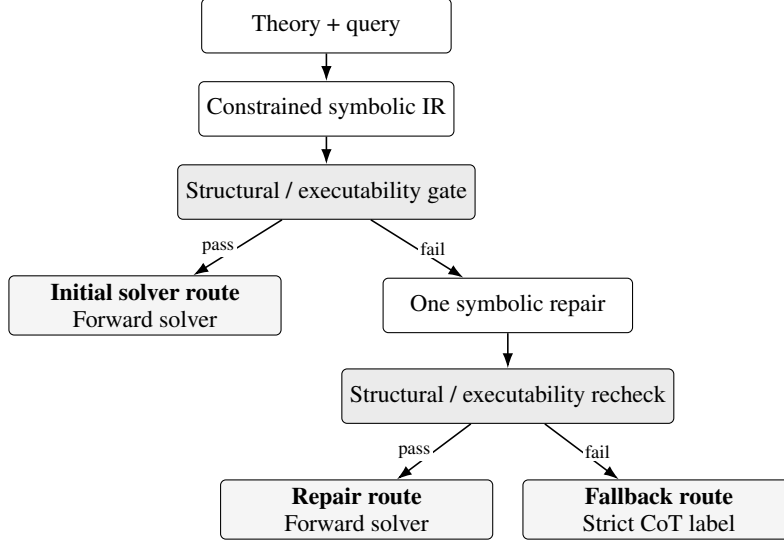


Figure 1: Full VGCF-2 branching policy. The gate tests structural validity and executability; it does not verify semantic equivalence to the source text.

3.2 Gate, repair, and fallback

The static gate first parses JSON, applies one documented normalization for a singleton rule premise, and checks the IR schema. Hard checks cover complete and unique source mapping, permitted identifiers, arity consistency, rule safety, ground facts and query, and the prohibition against inserting the query as a fact. Conservative lexical alignment and duplicate warnings are soft advisories; they do not alone block execution. The solver independently checks that the surviving representation can execute.

Full VGCF-2 uses the initial solver result only when the initial candidate is hard-valid and executable. Otherwise, a single repair prompt receives the original numbered input plus a bounded, severity-separated issue list. The repaired candidate is accepted only if it is hard-valid and executable. If that test fails, the method consumes a shared CoT response whose final label must match a strict anchored contract. Figure 1 summarizes this policy.

3.3 Compared methods

The evaluation contains exactly seven methods. *Direct* emits a label. *CoT* emits visible object-level reasoning and a strictly anchored label. *CoT-Refine* reviews and may revise the same question after an initial CoT response. *Constrained* uses only the shared initial formalization and solver, returning a method error when execution is unavailable. *Gate+CoT* executes a hard-valid initial formalization and otherwise uses the shared CoT fallback. *Repair-only* compares the shared initial and one repaired candidate using a fixed operational ordering, executes the selected candidate if possible, and has no CoT fallback. *Full VGCF-2* follows Figure 1. Sharing initial formalizations, repairs, and fallback responses makes route differences identifiable while avoiding artificial variation from duplicate calls. *Constrained* and *Repair-only* admit a selected candidate whenever it is solver-executable, whereas *Gate+CoT* and *Full VGCF-2* additionally require structural hard validity. *Repair-only* is therefore not simply *Full VGCF-2* without the CoT fallback; it also uses a weaker admission rule.

4 Experimental Design

Frozen evaluation. The Final evaluation uses a deterministic, label-balanced sample of 300 ProofWriter open-world questions, with at most one question selected from each of 300 unique theories. The local open-weight model identifier is qwen3.5-9b, from the Qwen3.5 family [Qwen Team, 2026]; the frozen decoding configuration uses temperature zero and reasoning_effort=none. The constrained and repair prompts each include one project-authored worked toy example to specify

the IR and output format; that example contains no ProofWriter item. Direct, CoT, and CoT-Refine are zero-shot apart from their output-contract instructions. CoT-Refine uses two logical model calls, whereas Full VGCF-2 can use up to three when both repair and fallback are required. These prompt and compute differences are part of the frozen comparison rather than post-hoc adjustments. All seven methods operate on the same question IDs. Development data, historically observed questions and theories, and the held-out Final manifest were separated by frozen quarantine rules; no Final-driven prompt or method tuning was permitted. The model and prompt hashes, sample manifest, decoding configuration, and source commit are frozen. An accuracy-independent preflight rule selected a completion budget of 3,000 tokens before the one-shot Final execution. The logical matrix contains 2,100 method results across the 300 questions, but these correspond to 1,315 fresh physical model requests: methods reuse the same newly generated constrained, CoT, and repair responses where the frozen design specifies sharing. This is deliberate within-run sharing, not reuse of stale responses. Because the Final sample is deliberately balanced across the three labels, its accuracy is for this designed evaluation distribution, not an estimate of population-weighted ProofWriter accuracy. Because our sample construction, model, and prompt configuration differ from prior ProofWriter studies, the absolute accuracies reported here should not be interpreted as controlled cross-paper comparisons with previously published results.

Dataset and model access. The ProofWriter V2020.12.3 dataset is available from the official AllenAI release archive¹, and Qwen3.5-9B is available from the official Qwen model card². Following the course submission guidance, dataset files and model weights are not included in the submission archive because of their size. The frozen local model identifier is qwen3.5-9b; we do not claim an exact hosted revision, quantization, or weight hash.

Metrics and failure handling. Overall accuracy counts a method error as incorrect. Macro-F1 is computed over the three task labels. Coverage is the fraction without a method error, and selective accuracy is accuracy among covered cases. These columns must be read together: a method can appear precise after abstaining on difficult cases. Because a method error contributes a false negative for its gold label without becoming a false positive for another task label, macro-F1 can also be inflated by abstention and should not be interpreted in isolation when error rates differ substantially.

Confirmatory hypotheses. Only two hypotheses are preregistered primary analyses. H1 tests whether Full VGCF-2 and CoT-Refine differ in paired correctness probability; H2 tests whether Full VGCF-2 and Gate+CoT differ. Both use paired outcomes on the same $n = 300$ questions, a two-sided exact McNemar test, Holm correction across the two-test family, and a paired bootstrap confidence interval for the accuracy difference. The reported difference is defined as Full VGCF-2 accuracy minus comparator accuracy, so positive values favor Full. All other breakdowns and diagnostics in this paper are secondary descriptive, mechanism-descriptive, archival, or explicitly labeled post-hoc exploratory or measurement-validity analyses.

5 Results

5.1 Overall Performance

Table 1 reports the frozen secondary descriptive method metrics. Full VGCF-2 is highest on accuracy and macro-F1, with 1 method error and 99.67% coverage. The Constrained and Repair-only rows also illustrate why coverage matters: their selective accuracy is high relative to overall accuracy, but many questions have no returned label.

5.2 Preregistered Primary Comparisons

Both preregistered two-sided null hypotheses are rejected after Holm adjustment, and both observed accuracy differences favor Full VGCF-2 (Table 2). For H1, Full VGCF-2 exceeds CoT-Refine by 8.67 percentage points, with 41 wins and 15 losses among discordant pairs; the paired interval is [4.00, 13.33] points. For H2, Full VGCF-2 exceeds Gate+CoT by 2.67 points, with 10 wins and

¹<https://aristo-data-public.s3.amazonaws.com/proofwriter/proofwriter-dataset-V2020.12.3.zip>

²<https://huggingface.co/Qwen/Qwen3.5-9B>

Table 1: Final method performance on the same 300 questions. Accuracy, macro-F1, coverage, selective accuracy, and method errors are secondary descriptive metrics; method errors count as incorrect in overall accuracy.

Method	Accuracy	Macro-F1	Coverage	Selective acc.	Errors
Direct	38.00%	37.67%	100.00%	38.00%	0
CoT	78.67%	79.88%	98.33%	80.00%	5
CoT-Refine	81.67%	82.55%	98.67%	82.77%	4
Constrained	57.67%	70.70%	63.33%	91.05%	110
Gate+CoT	87.67%	88.23%	99.33%	88.26%	2
Repair-only	74.67%	82.42%	81.33%	91.80%	56
Full VGCF-2	90.33%	90.67%	99.67%	90.64%	1

2 losses; its paired interval is [0.67, 5.00] points. The Holm-adjusted values are $p = 0.001371$ and $p = 0.038574$, respectively. These results support the full composition relative to the frozen CoT-Refine configuration and, within this pipeline and admission policy, a smaller incremental benefit from including the one-shot repair route relative to gate-plus-fallback.

Table 2: Preregistered primary paired tests. Δ is Full VGCF-2 accuracy minus comparator accuracy; W/L counts discordant Full wins and losses. Both two-sided null hypotheses are rejected after Holm adjustment; the observed accuracy differences favor Full VGCF-2 in both comparisons.

Hyp.	Comparison	Accuracy	Δ pp	W/L	Raw p	Holm p	95% CI (pp)
H1	Full VGCF-2 vs. CoT-Refine	90.33%/81.67%	8.67	41/15	0.000686	0.001371	[4.00, 13.33]
H2	Full VGCF-2 vs. Gate+CoT	90.33%/87.67%	2.67	10/2	0.038574	0.038574	[0.67, 5.00]

5.3 Routing and Repair Mechanisms

The mechanism-descriptive route counts in Table 3 locate, but do not causally prove, the source of the gain. Full VGCF-2 solved 185 cases from the initial representation, 54 after accepted repair, and 60 through CoT fallback. Gate+CoT sends more cases directly to fallback because it has no repair stage. Across 115 shared repair attempts, the frozen operational diagnostic records 57 improvements and 3 regressions. These are secondary route-quality descriptions under the fixed selection policy.

The frozen H2 discordant-case analysis provides a second mechanism view. Full has 10 gains and 2 losses relative to Gate+CoT; in every discordant case, Full used the repaired-solver route while Gate+CoT followed its no-repair alternative (CoT fallback or method error). This pattern locates the paired differences on the repair route under the frozen design. This is a secondary mechanism observation rather than a confirmatory analysis.

5.4 Formalization Quality and Symbolic Coverage

Table 4 reports complementary initial-formalization diagnostics rather than a single nested funnel. Strict and normalized parsing form separate tracks, and structural hard validity is a stricter admission condition than solver executability, so the counts need not decrease monotonically. On the full 300-question cohort, 288/300 initial responses parse as JSON, 192/300 satisfy the normalized schema, 185/300 pass the structural hard validator, and 190/300 are solver-executable after normalization. The symbolic-only Constrained method is correct on 173 of its 190 covered cases (91.05% selective accuracy), but it covers only 190/300 questions. Symbolic execution can therefore be highly accurate when a usable representation exists, while incomplete formalization coverage still limits an all-symbolic policy.

Semantic faithfulness is a separate representational limitation. Among the 192 normalized-schema-valid representations, only 100/192 meet the stricter renaming-invariant semantic-exact diagnostic. Structural acceptance and solver executability establish operational usability, not semantic equivalence to the natural-language theory.

Table 3: Secondary mechanism description of realized routes and correct answers. Route counts partition each method’s Final sample.

Full route	n	Correct	Gate+CoT route	n	Correct
Initial solver	185	169	Initial solver	185	169
Repaired solver	54	52	CoT fallback	113	94
CoT fallback	60	50	Method error	2	0
Method error	1	0	–	–	–

Table 4: Initial-formalization diagnostics. Strict and normalized rows belong to separate diagnostic tracks; structural hard validity and solver executability are distinct conditions rather than nested stages.

Initial-formalization diagnostic	Count	Denominator
JSON parse valid	288	300
Strict schema valid	180	300
Strict solver executable	178	300
Normalized schema valid	192	300
Hard-validator valid	185	300
Normalized solver executable	190	300
Renaming-invariant semantic exact	100	192

An explicitly post-hoc matched admission ablation reused the frozen responses with no new model calls and relaxed hard-validity admission to solver executability alone. Only five Gate+CoT routes and seven Full routes changed; accuracy moved from 263/300 to 264/300 for Gate+CoT and from 271/300 to 270/300 for Full. This provides no consistent accuracy advantage for hard validity beyond executability in this frozen sample.

5.5 Secondary Semantic-Fidelity Analysis

A post-hoc AI-assisted review compared each visible rationale’s semantic conclusion with its machine-readable anchor and the deterministic parser. For initial CoT, 295/300 responses were contract-valid and 292/295 of those (98.98%) had semantic-anchor agreement. For CoT-Refine, 296/300 were contract-valid and all 296/296 (100.00%) agreed. Anchor-parser fidelity was 295/295 for initial CoT and 296/296 for CoT-Refine.

These measurements test rationale-to-output consistency, not whether the reasoning is logically correct. The three initial contract-valid mismatches have rationales semantically consistent with Unknown but a False machine anchor and parser output. They remain part of the frozen record and were not rescored; Appendix C records their case context.

5.6 Post-Hoc Defect Decisiveness

To examine whether global semantic inexactness implies downstream answer impact, we performed a no-new-model-inference, evaluator-only counterfactual analysis on the 115 semantic-inexact selected IRs among Full VGCF-2’s 239 symbolic-route cases. For each source mismatch that could be aligned under the established renaming-invariant mapping, we replaced exactly one predicted source element with its counterpart in the evaluator-only ProofWriter reference program, left all other IR elements unchanged, and reran the deterministic solver. The reference program was used only for this post-hoc evaluator analysis after the frozen Final execution; it was never exposed through the model-input gateway. Of 248 identified mismatch candidates, 216 were testable. Only 22/216 (10.2%) single-element corrections changed the solver label, and 14/216 (6.5%) individually corrected an originally wrong answer.

A complementary non-deployable oracle shows why global inexactness is not itself a safe routing criterion. Replacing every renaming-invariant-inexact symbolic-route result with its already-frozen CoT alternative reduced whole-Full accuracy from 271/300 to 265/300 (13 gains and 19 losses). This post-hoc result uses evaluator-only semantic labels and is diagnostic rather than a deployable policy.

At the case level, 14/17 (82.4%) wrong semantic-inexact cases with at least one tested defect contained an individually corrective defect, whereas 94/96 (97.9%) originally correct inexact cases with at least one tested defect preserved their answer under every tested single-element correction. As a methodological sanity check, for all 108 fully patchable cases, correcting all identified defects reconstructed a reference-equivalent program and recovered the frozen gold label. These post-hoc interventions do not alter any frozen prediction or define a deployable verifier; Appendix G reports the method, coverage, and interaction caveats.

6 Discussion

Under the frozen prompt and decoding configuration, Full VGCF-2 outperformed CoT-Refine. This is not a strategy-only or compute-matched contrast: the constrained and repair prompts include one project-authored format demonstration, the natural-language prompts are zero-shot, and Full can use an additional repair/fallback call path. H1 should therefore be read as evidence for the tested full configuration rather than as a general claim that symbolic routing is intrinsically superior to natural-language self-review. Within that configuration, structural validation and deterministic execution expose failure signals such as JSON validity, source coverage, rule safety, executability, and proof tracing that natural-language self-review does not directly provide. Conversely, the full pipeline does not assume every question should be solved symbolically: fallback preserves coverage when a candidate remains inadmissible.

H2 identifies a smaller, statistically supported design contrast within the frozen pipeline. Gate+CoT and Full VGCF-2 share their initial symbolic response, initial gate decision, and fallback response; their difference is the bounded repair opportunity. The 2.67-point effect rests on 12 discordant pairs (10 Full wins and 2 losses), so we treat it as a modest sample-specific effect rather than as evidence of broad practical importance. The discordant-case pattern is consistent with an incremental benefit from the one-shot repair route under this admission policy, but it does not imply that arbitrary repeated repair is beneficial. Repair acceptance remains strict, and a model-generated edit can still degrade a candidate.

The matched admission ablation therefore provides no consistent accuracy advantage from structural hard validity beyond solver executability. We interpret hard validity here primarily as a contract and provenance safeguard rather than as an isolated accuracy driver.

The formalization diagnostics clarify the system’s selective trade-off. Deterministic solving is effective once a usable IR exists, but the gap between executable and semantic-exact representations shows that representation fidelity remains imperfect. The complementary non-deployable oracle shows that global semantic inexactness is too coarse to serve as a safe fallback or routing criterion under the frozen alternatives. Section 5.6 sharpens this point: most tested local semantic defects did not change the solver label under isolated correction, even though most testable wrong inexact cases contained at least one individually corrective defect. Structural admission and output-contract validity are not semantic certification, but neither should every semantic mismatch be treated as equally consequential. The semantic-fidelity review adds a separate measurement point: exact anchor parsing can coexist with a rationale whose stated conclusion means something else.

This distinction suggests a narrower direction for future verification: prioritize whether a source-to-IR defect is query- or proof-relevant and whether correcting it can change derivability, rather than auditing global faithfulness alone. The present counterfactual analysis is evaluator-only and cannot provide such a deployable selector. A prospective extension should therefore define a gold-blind relevance mechanism before evaluation and compare targeted verification with broader global semantic checks on an untouched sample. Counterfactual defect localization can also serve as an evaluator for whether such a mechanism is focusing computation on answer-relevant errors rather than merely detecting any representation mismatch.

7 Limitations

The evaluation uses one frozen sample from a synthetic benchmark and one frozen model configuration. Results may not transfer to less regular language, broader first-order logic, or other model families. The IR supports only signed unary and binary atoms with forward rules, so the method does not cover arbitrary quantification, defeasibility, or commonsense premise generation.

The gate is intentionally structural. Lexical advisories are conservative, and hard validity plus solver executability cannot establish meaning preservation. Exact formalization diagnostics are representation-sensitive even after renaming invariance, and the post-hoc AI-assisted semantic-fidelity review measures rationale-to-output consistency rather than reasoning correctness. The defect-decisiveness analysis is likewise post-hoc and evaluator-only. Coverage is incomplete because 32/248 identified mismatch candidates were not testable under the established inverse renaming map (Appendix G); a single-element intervention establishes only the consequence of that tested counterfactual, not global benignity or causal sufficiency outside the tested IR.

Coverage and accuracy are also tied to the frozen output contracts, prompts, and compute allocation. Method errors are counted as incorrect rather than silently recovered. The constrained and repair prompts contain one project-authored format demonstration, whereas Direct, CoT, and CoT-Refine are zero-shot, and Full can use more logical model calls than CoT-Refine. Native model reasoning was disabled uniformly with `reasoning_effort=none`, but we did not test whether enabling it would affect the natural-language and symbolic routes equally. H1’s magnitude should therefore be interpreted as specific to this frozen prompt and decoding configuration.

ProofWriter has been publicly available since 2021, and we cannot determine whether the evaluated model encountered it during pretraining. Absolute accuracies may therefore benefit from prior benchmark exposure. The two primary comparisons are paired within the same model and sample, which reduces but does not eliminate this concern because exposure could interact differently with different prompt and reasoning formats.

8 Conclusion

VGCF-2 combines model-based language interpretation with deterministic validation and deduction, while retaining a natural-language escape route when symbolic execution is not admissible. On the frozen ProofWriter evaluation, the full gate-repair-fallback system outperforms both CoT-Refine and the otherwise shared Gate+CoT method in the two preregistered paired tests. The broader lesson is narrower than “verification solves reasoning”: structural checks can support more auditable selective execution, while semantic faithfulness remains a separate problem.

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A Reproducibility

The reported Final run is tied to frozen source, configuration, sample, and result identities. Existing outputs can be reverified offline with `py -3.12 scripts/post_final/materialize_final_v2_3_1.py -project-root . -verify-only`, and the paper-number bindings can be checked with `py -3.12 paper/tools/verify_frozen_sources.py -project-root ..` Neither verification requires a model server or network access.

Re-running the frozen Final evaluation itself uses `py -3.12 scripts/run_experiment.py -project-root . -config configs/final_test_v2_3_1.json` and requires the configured local OpenAI-compatible LM Studio server hosting `qwen3.5-9b`. The archived run issued 1,315 physical requests totaling 1,375,249 tokens, with approximately 7.0 hours of summed request latency and 7.1 hours of observed wall-clock time. Archived run artifacts preserve the raw model responses.

B Development-to-Final Descriptive Check

Table 5 is a descriptive consistency check only, not a preregistered generalization hypothesis. Development results informed the frozen protocol; the one-shot Final results remained held out from tuning.

Table 5: Development and Final accuracy from the frozen descriptive comparison. No inferential claim is attached to these differences.

Method	Dev accuracy	Final accuracy
Direct	38.00%	38.00%
CoT	77.33%	78.67%
CoT-Refine	84.00%	81.67%
Constrained	54.00%	57.67%
Gate+CoT	84.00%	87.67%
Repair-only	69.33%	74.67%
Full VGCF-2	88.00%	90.33%

C Semantic-Fidelity Case Context

The post-hoc AI-assisted Final review recorded 18 semantic revisions and 3 semantic-anchor mismatches. These are measurement-validity evidence only. The mismatch identifiers are `RelNeg-OWA-D5-404:Q22`, `RelNoneg-OWA-D5-142:Q11`, and `RelNoneg-OWA-D5-826:Q14`. In every case the visible rationale concluded Unknown while the machine-readable anchor and parser output were False. Both Gate+CoT and Full VGCF-2 used the initial-solver route on all three, so none of the mismatched anchors was consumed as fallback. They do not change any method output or score; this route observation is post-hoc interpretability evidence.

D Secondary Label and Depth Slices

Table 6 shows Full VGCF-2 only. These slices were not preregistered hypotheses and should not be treated as independent tests. Label rows are balanced by the frozen sampling design. Depth and gold label are not independent in this sample: Unknown items are concentrated at lower depths, whereas higher depths are dominated by True and False items. The depth rows therefore should not be read as a pure reasoning-difficulty gradient.

Table 6: Secondary descriptive Full VGCF-2 results by gold label and proof depth.

Slice	n	Correct	Accuracy	Coverage
Gold True	100	89	89.00%	100.00%
Gold False	100	87	87.00%	100.00%
Gold Unknown	100	95	95.00%	99.00%
Depth 0	73	71	97.26%	98.63%
Depth 1	65	64	98.46%	100.00%
Depth 2	42	38	90.48%	100.00%
Depth 3	44	37	84.09%	100.00%
Depth 4	36	31	86.11%	100.00%
Depth 5	40	30	75.00%	100.00%

E Method Contracts and Failure Semantics

Direct classification accepts only a task label. CoT and CoT-Refine require a visible reasoning or review section followed by exactly one final anchored label as the last non-empty line. Constrained formalization accepts only one IR JSON object. Any parse failure, rejected route, solver inconsistency, or invalid fallback contract is explicit; the evaluator never silently maps it to Unknown.

The proof solver preserves signed atoms and shared substitutions across conjunctive premises. It applies implications only forward and never inserts the query among the facts. Consequently, failure to derive a query does not yield False: the explicit complement must be derivable, otherwise the answer is Unknown. These implementation invariants are part of the frozen source lock, not conclusions inferred from Final labels.

F Illustrative Post-Hoc Cases

These cases were selected deterministically by frozen manifest order within three predeclared route/outcome categories after the main analysis. They are illustrative only and provide no additional statistical evidence.

Repair gain. For AttNoneg-OWA-D5-513:Q7, the initial formalization encoded several implication sentences as facts, so it failed schema validation and Gate+CoT fell back to an incorrect Unknown prediction. One repair converted those entries into executable rules. The accepted program derived $\text{smart}(\text{Fiona}) \rightarrow \text{cold}(\text{Fiona}) \rightarrow \text{kind}(\text{Fiona}) \rightarrow \text{nice}(\text{Fiona})$, yielding the correct label True.

Fallback after structural rejection. For AttNoneg-OWA-D5-336:Q7, repair corrected a malformed rule and made the program solver-executable, but the repaired IR still omitted one source sentence. Full VGCF-2 therefore rejected it under structural hard validity and used the frozen CoT fallback, which returned the correct label True. In the post-hoc matched admission ablation, executability-only admission would also have returned True from the repaired solver. The case therefore illustrates that hard validity enforces the source-coverage contract beyond executability, not an accuracy gain in this instance.

Repair loss. For RelNeg-OWA-D5-552:Q8, the initial formalization already represented one source rule with a variable where the source sentence specified the bald-eagle constant. Repair corrected other structural issues and produced a hard-valid and executable program while preserving that semantic defect. The resulting solver prediction was Unknown and incorrect, whereas Gate+CoT’s frozen fallback returned the correct False label. This case illustrates that repair can make a candidate structurally admissible without correcting a pre-existing semantic error.

G Post-Hoc Defect Decisiveness Analysis

The analysis in Section 5.6 is an evaluator-only counterfactual diagnostic rather than a deployable component of VGCF-2. A source element is aligned by source identifier using the existing kind-sensitive, alpha-invariant, premise-order-invariant comparison and the selected IR’s established predicted-to-reference constant mapping. For each testable mismatch, exactly one predicted source

element is replaced by its reference counterpart under the inverse established mapping; all unrelated IR elements remain unchanged before the deterministic solver is rerun. No identity, lexical, or guessed constant mapping is introduced when the inverse mapping is unavailable.

Table 7 summarizes the resulting coverage and counterfactual outcomes. Full, partial, and no analysis coverage mean that all, some, or none of a case’s identified mismatch candidates were testable under the established inverse mapping.

Table 7: Post-hoc evaluator-only defect-decisiveness diagnostics. Denominators reflect testable defects or cases.

Diagnostic	Result
Semantic-inexact symbolic-route cases	115
Full / partial / no analysis coverage	108 / 5 / 2
Tested / identified mismatch candidates	216 / 248
Single-patch label-changing defects	22/216 (10.2%)
Individually corrective defects	14/216 (6.5%)
Wrong tested cases with ≥ 1 corrective defect	14/17 (82.4%)
Correct tested cases with all patches preserving answer	94/96 (97.9%)
All-defect reference-equivalence / gold-label recovery	108/108

A defect is called individually corrective only when its isolated correction changes an originally wrong symbolic answer to the frozen gold label. An unchanged answer under a single patch is only single-patch nondecisiveness; it does not establish that the defect is globally benign. Conversely, changing an originally correct answer after correcting one element can expose interactions among compensating defects and is not interpreted as evidence that the reference correction is harmful.

One fully analyzable wrong case, `Re1Noneg-DWA-D5-462:Q10`, illustrates interaction among defects: none of its three tested single-element corrections was individually sufficient, whereas correcting all identified defects reconstructed the reference program and recovered the gold label. Across all 108 fully patchable cases, the all-defect intervention reconstructed a reference-equivalent program and recovered the frozen gold label, providing a methodological sanity check for the patching procedure.

Coverage is not complete. Thirty-two of 248 identified mismatch candidates were left untested because required reference constants had no inverse under the established renaming map; no replacement was guessed for those cases. The resulting counterfactuals support studying answer-relevant semantic defects as a sharper exploratory object than global semantic inexactness alone, but they establish neither a deployable verifier nor causality beyond the exact interventions tested here.